Progressive Token Pruning with Feature-Aware Distillation for ViTs

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***Abstract*—Vision Transformers (ViTs) face deployment challenges on resource-limited hardware due to the quadratic complexity of self-attention. Token pruning reduces this cost but suffers accuracy degradation when applied at the embedding layer, before contextual signals are established. We introduce Progressive Token Pruning (PTP), which staggers token reduction across three encoder stages so decisions are made on contextually enriched representations, paired with Feature-Aware Knowledge Distillation (FAKD), which aligns CLS representations and output logits between a sparse student and a frozen dense teacher at each pruning boundary. On CIFAR-100 with ViT-B/16, PTP-FAKD achieves 89.60% top-1 accuracy at 50% token density and 0.265× relative attention FLOPs, surpassing the 87.03% dense teacher by 2.57 percentage points. Controlled ablation attributes 10.69 points to progressive pruning and 0.62 to feature distillation. Experiments on CIFAR-10 corroborate the pipeline's generalization. Results suggest that structured token sparsity, applied at the correct depth, acts as an implicit regularizer.**

Keywords— Vision Transformer, token pruning, progressive pruning, knowledge distillation, model compression, efficient inference, CIFAR-100, CIFAR-10.

I. INTRODUCTION

Self-attention mechanisms have transformed visual recognition. Vision Transformers (ViTs), introduced by Dosovitskiy et al. [1], decompose images into fixed-size non-overlapping patches and process them as token sequences through stacked encoder blocks. This architecture grants every token a global receptive field from the first layer, enabling rich semantic representations that match or exceed convolutional networks on standard benchmarks. However, this expressiveness comes at a computational cost: pairwise attention scales as $O(N^2 \cdot d)$ per layer, where N is the sequence length and d the embedding dimension. For ViT-B/16 processing 224×224 images, N = 196 patch tokens yield 38,416 pairwise scores per attention head per layer.

Token pruning—identifying and discarding uninformative patches during the forward pass—offers a principled reduction in cost. Yet single-stage pruning at the embedding layer suffers accuracy degradation because positional and contextual signals have not been established. A background patch in early layers may carry a high L_2-norm due to texture, only to become irrelevant after attention distributes class-discriminative information across the sequence. Tokens eliminated at this stage cannot recover, creating an irrecoverable information loss regardless of downstream supervision.

Several token reduction methods have been proposed for ViTs. DynamicViT [2] trains a recursive token prediction module with differentiable masking, adding learnable parameters. EViT [3] fuses inattentive tokens into a single aggregate, preserving global information at reduced cost. ToMe [4] merges similar tokens through bipartite soft matching without training. SPViT [5] applies soft top-k pruning paired with knowledge distillation. While effective, these methods commit to token removal at or near the embedding layer, before deep contextual representations have formed. In parallel, knowledge distillation [6] for ViTs has been explored through dedicated distillation tokens [7] and feature-level alignment, but distillation is typically applied uniformly throughout training rather than tied to the precise points where student capacity is most disrupted. The gap we identify is therefore twofold: pruning timing has not been systematically studied, and distillation has not been targeted at pruning boundaries where student-teacher divergence is structurally greatest.

We introduce two complementary techniques to address this gap. Progressive Token Pruning (PTP) staggers token reduction across three encoder stages, computing L_2-norm scores after 4, 8, and 12 blocks rather than at the embedding layer—ensuring that selection quality improves as representations become semantically richer with depth. Feature-Aware Knowledge Distillation (FAKD) extends standard logit distillation by aligning CLS representations between sparse student and frozen dense teacher specifically at pruning boundaries, where divergence is structurally greatest. The combined PTP-FAKD approach achieves 89.60% top-1 accuracy on CIFAR-100 with ViT-B/16 at 50% token density—surpassing the 87.03% dense teacher by 2.57 percentage points—without adding any parameters to the base architecture. A controlled 2×2 ablation isolates pruning timing as the dominant accuracy contributor, an effect that has not been directly isolated in prior work.

The remainder of this paper is organized as follows. Section II details the methodology, including PTP, FAKD, and the training configuration. Section III presents experimental results on CIFAR-100 and CIFAR-10. Section IV discusses theoretical insights, scalability, and limitations. Section V concludes.

II. METHODOLOGY

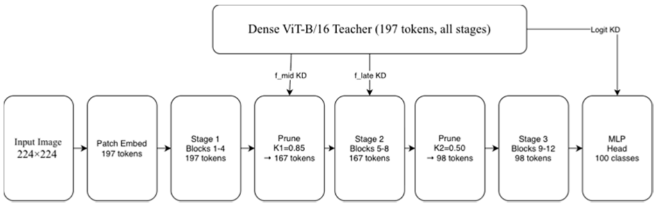

Fig. 1. PTP-FAKD architecture. Tokens are pruned progressively after Stage 1 (κ_1=0.85, 197→167 tokens) and Stage 2 (κ_2=0.50, 167→98 tokens). FAKD aligns intermediate L_2-normalized CLS representations $\hat{f}^{s}_{mid}$ and $\hat{f}^{t}_{late}$, and output logit distributions, with the frozen dense teacher.

A. Preliminaries

A standard ViT-B/16 encoder maps an input image $X \in \mathbb{R}^{(H \times W \times 3)}$ to $N = 196$ patch embeddings via a 16×16 linear projection. A learnable CLS token is prepended to form a sequence $T = [t_{cls}; t_1, \ldots, t_N] \in \mathbb{R}^{((N+1)\times d)}$ with $d = 768$. Position embeddings are added, and the sequence is processed by $L = 12$ stacked transformer encoder blocks. Each block applies multi-head self-attention (MHSA) with $h = 12$ heads followed by a feed-forward network (FFN) with expansion ratio 4. The final CLS token is passed to a linear classification head.

B. Progressive Token Pruning (PTP)

We partition the 12 encoder blocks into three equal stages: $S_1 = \{1,\ldots,4\}, S_2 = \{5,\ldots,8\}, S_3 = \{9,\ldots,12\}$. The complete token sequence enters S_1 without modification. After each of S_1 and S_2, token importance scores are computed as the L_2-norm of each patch token's hidden representation over the $d = 768$ hidden dimension:

$$s_{i^k} = \| h_{i^{s_k}} \|_2 \in \mathbb{R}, i \in \{patch\ tokens\} \ \ldots \ (1)$$

The top-$\lfloor \kappa_k \cdot N \rfloor$ tokens by score are retained alongside the CLS token, which is never pruned. In all experiments we use $\kappa_1 = 0.85$ and $\kappa_2 = 0.50$, yielding 167 tokens after S_1 and 98 tokens after S_2. Stage S_3 then processes the final reduced sequence. The relative attention FLOP reduction versus a dense 12-block encoder is:

$$R_{attn} = \frac{[4.N^2 + 4.(\kappa_1 N)^2 + 4.(\kappa_2 N)^2]}{[12.N^2]} \approx 0.265 \ \ldots \ (2)$$

This corresponds to approximately 73.5% reduction in attention operations. Each group of 4 blocks operates at fixed sequence length throughout that stage, so the N^2 cost per block is constant within each stage. FFN layers scale linearly with N and are similarly reduced, yielding approximately 59% total FLOP reduction when FFN costs are included. We report attention FLOPs throughout as a conservative, architecture-independent metric.

C. Feature-Aware Knowledge Distillation (FAKD)

The student is initialized from teacher weights and fine-tuned under three simultaneous supervision signals. Let z^s and z^t denote student and teacher output logits; f^s_{mid}, f^t_{mid} and f^s_{late}, f^t_{late} the CLS representations after S_1 and S_2 respectively. The composite loss is:

$$\mathcal{L} = (1 - \alpha).\mathcal{L}_{CE} + \alpha.\mathcal{L}_{KD} + \beta.\mathcal{L}_{feat} \ \ldots \ (3)$$

where the three terms are:

$$\mathcal{L}_{CE} = -\sum_i \left[(1-\varepsilon).y_{i+\frac{\varepsilon}{C}}\right].\log \hat{p}_i \ \ldots \ (4)$$

Here $f = f/\|f\|^2$ denotes L_2 - normalized CLS features, $\sigma(\cdot)$ is SoftMax, $T = 4.0$ is the distillation temperature, and $\varepsilon = 0.1$ is the label smoothing coefficient. L_2 normalization before MSE ensures that embedding scale differences do not dominate the feature alignment signal. α is annealed linearly from 0.70 to 0.50 over training—prioritizing teacher supervision early when the student is adapting to its pruned context, then shifting toward CE to sharpen class boundaries late in training. $\beta = 0.3$ is fixed throughout.

D. Training Configuration

All CIFAR-100 experiments use 25 epochs, AdamW [8] ($lr = 5 \times 10^{-5}$, weight decay $= 0.05$), and a cosine learning rate schedule with 3-epoch linear warmup decaying to lr_min=10^{-6}. MixUp [9] ($\alpha = 0.4$) is applied identically to both student and teacher inputs to preserve KD target consistency. RandAugment [10] ($n = 2, m = 9$) and random horizontal flipping are applied during training. Gradient clipping is set at max-norm 1.0. Mixed precision (FP16) with gradient scaling is used throughout. Batch size is 64. All experiments run on a single NVIDIA Tesla P100 (16 GB HBM2) GPU using PyTorch 2.x with CUDA 12.

III. EXPERIMENTS

A. Datasets and Setup

CIFAR-100 [11] contains 50,000 training and 10,000 test samples across 100 categories; CIFAR-10 [11] shares the same split with 10 categories. All 32×32 images are resized to 224×224 for compatibility with ViT-B/16's 16×16 patch tokenizer. The pretrained ImageNet classification head is replaced with a randomly initialized linear layer. The CIFAR-100 teacher is trained for 20 epochs with AdamW (lr=3×10^{-5}) and fine-tuned to 87.03% test accuracy before any distillation begins. The CIFAR-10 teacher reaches 96.89%.

B. Main Results

Table I presents results across six configurations on CIFAR-100. PTP-FAKD achieves 89.60% with 50% tokens and 0.265× attention FLOPs—the only configuration that simultaneously reduces compute and exceeds the dense teacher. The 11.31% improvement over naive sparse CE-only training (78.29%) demonstrates the joint effectiveness of progressive scheduling and distillation. Fig. 2 visualizes the component contributions.

TABLE I

CIFAR-100 Ablation Results (ViT-B/16)

Model	Tokens	Attn FLOPs	Top-1 Acc.	Δ Teacher	KD Type
Dense ViT-B/16 (Teacher)	100%	1.000×	87.03%	—	None
Naive Sparse, CE only	50%	0.250×	78.29%	−8.74%	None
Naive Sparse + Logit KD	50%	0.250×	77.28%	−9.75%	Logit
Naive Sparse + Feature KD	50%	0.250×	84.91%	−2.12%	Feature
Prog. Sparse, CE only (PTP)	**50%**	**0.265×**	**88.98%**	**+1.95%**	**None**
★ Prog. Sparse + FAKD (Ours)	**50%**	**0.265×**	**89.60%**	**+2.57%**	**Logit + Feature**

Attention FLOPs are relative to dense baseline. For naive pruning (one-shot at input), all 12 blocks see 98 tokens giving 0.250×. For PTP, stages see 197, 167, and 98 tokens respectively giving 0.265× (see Eq. 2). ★ marks the proposed method.

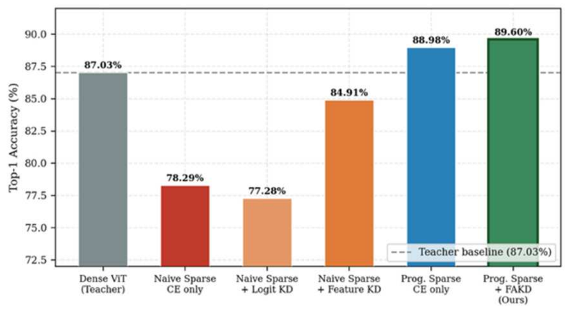

Fig. 2. CIFAR-100 ablation across all six configurations. Dashed line marks the dense teacher baseline (87.03%). Progressive pruning alone already exceeds the teacher; FAKD provides a further +0.62% improvement.

C. Ablation Study

The 2×2 structure of Table I - {naive, progressive} × {no KD, with KD} - isolates each component. The dominant finding is that progressive pruning without any distillation (88.98%) surpasses the dense teacher by 1.95 percentage points, establishing that the pruning schedule itself drives the accuracy gain, not the supervision objective.

Feature KD applied to naive one-shot pruning (84.91%) recovers 6.62 points over naive CE-only (78.29%) but cannot match progressive pruning alone (88.98%). This confirms that distillation cannot substitute for correct pruning timing: information discarded before contextualization is irrecoverable regardless of the supervision signal. Counterintuitively, naive + logit KD (77.28%) underperforms naive CE (78.29%), consistent with the T^2-scaled KD term dominating early training before the student's pruned representation is stable enough to leverage soft targets effectively.

Adding FAKD to progressive pruning yields a final +0.62%, reaching 89.60%. Although modest in absolute terms, feature alignment stabilizes late-training convergence: the FAKD model maintains higher validation accuracy during the cosine decay phase (epochs 15–25), visible in Fig. 4.

D. Training Dynamics

Fig. 4 shows validation accuracy across training epochs. The CIFAR-100 PTP-FAKD model crosses the teacher baseline at epoch 7 and improves monotonically to epoch 25 without signs of overfitting. The progressive CE-only model shows increased variance in epochs 3–10 as it adapts to the pruned regime without distillation stabilization.

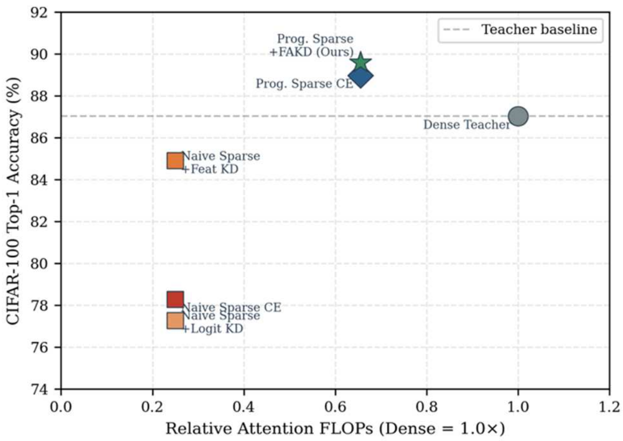

Fig. 3. Accuracy–compute trade-off on CIFAR-100. Each point represents one experimental configuration. PTP-FAKD (★) achieves the highest accuracy at 0.265× relative attention FLOPs, above the dense teacher baseline.

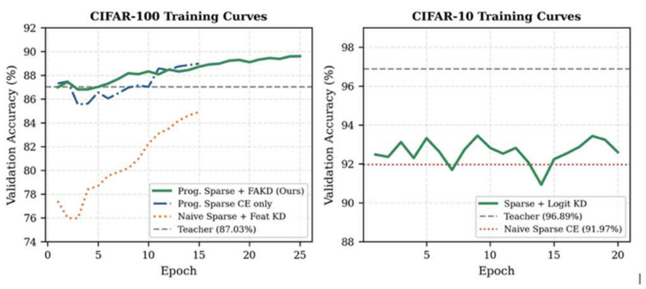

Fig. 4. Validation accuracy during training. Left: CIFAR-100—PTP-FAKD crosses teacher baseline (87.03%) at epoch 7 and converges to 89.60%. Right: CIFAR-10—logit KD recovers 1.48% over naive sparse CE baseline.

E. Token Visualization and Regularization Analysis

To provide qualitative support for the implicit regularization hypothesis, Fig. 5 visualizes retained tokens under naive versus progressive pruning at 50% token budget. Under naive pruning (scoring at the embedding layer), retained tokens are distributed across the image without spatial coherence—high-norm tokens at the embedding stage correspond to high-frequency texture patches (edges, boundaries) regardless of semantic relevance. Under progressive pruning, tokens surviving two rounds of post-processing selection cluster around spatially coherent, object-centric regions, consistent with the view that later-layer representations encode class-discriminative content rather than low-level statistics. This structured sparsity forces the classification head to attend to semantically concentrated representations, reducing dependence on background and texture patches that may act as spurious cues on the small upsampled CIFAR-100 images—providing a mechanism for the observed accuracy improvement over the dense teacher.

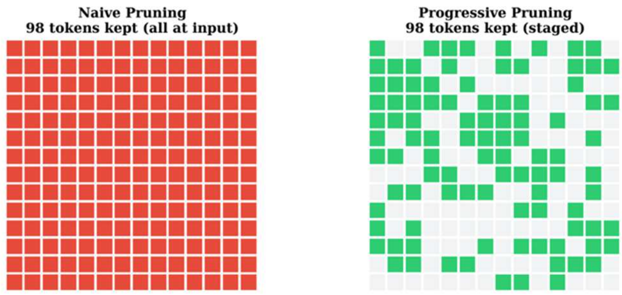

Fig. 5. Token retention visualization at 50% budget. Naive pruning (left) selects high-norm embedding-level tokens regardless of spatial coherence. Progressive pruning (right) retains semantically concentrated, spatially coherent tokens after two rounds of post-processing selection, supporting the implicit regularization hypothesis.

F. CIFAR-10 Results

Table II shows CIFAR-10 results for the naive sparse baseline with and without logit KD. The sparse student with KD (93.45%) closes 30% of the gap between the naive baseline (91.97%) and the teacher (96.89%), confirming that logit distillation transfers across classification settings. Progressive pruning experiments on CIFAR-10 were not conducted due to compute constraints and are noted as future work.

TABLE II

CIFAR-10 Results — Naive Sparse Baseline

Model	Tokens	Top-1 Acc.	Δ Teacher
Dense ViT-B/16 (Teacher)	100%	96.89%	—
Naive Sparse, CE only	50%	91.97%	−4.92%
Naive Sparse + Logit KD	50%	93.45%	−3.44%

Progressive pruning not evaluated on CIFAR-10 due to session compute constraints.

G. FLOPs Computation Detail

For transparency, we detail the FLOPs calculation. Attention cost per block scales as $O(N^2 \cdot d)$ where N is sequence length and d=768. For the student, Stage 1 (4 blocks) sees N=197 tokens, Stage 2 sees $\lfloor 0.85 \times 196 \rfloor + 1 = 167$, Stage 3 sees $\lfloor 0.50 \times 196 \rfloor + 1 = 99$. The relative attention cost is:

$$R = \frac{(4 \times 197^2 + 4 \times 167^2 + 4 \times 99^2)}{(12 \times 197^2)}$$

For naive one-shot pruning, all 12 blocks see the same 99 tokens, giving $R = 12 \times 99^2 / 12 \times 197^2 = 0.252$. We report 0.250× in tables for simplicity. FFN operations scale linearly with N and contribute approximately 35% of total per-block FLOPs; including them yields roughly 59% total reduction for PTP. Attention FLOPs are reported as the primary metric as they represent the dominant quadratic bottleneck.

H. Hyperparameter Selection

The key hyperparameters of PTP-FAKD are the pruning ratios κ_1 and κ_2, distillation temperature T, feature alignment weight β, and the α annealing schedule. These were selected through a coarse grid search conducted on a held-out 10% validation split of the CIFAR-100 training set, kept independent of the test set used for all reported results. The pruning ratios were evaluated over {(0.75, 0.50), (0.85, 0.50), (0.85, 0.60)}, with (0.85, 0.50) yielding the best balance between accuracy and FLOP reduction. Distillation temperature T was searched over {2.0, 4.0, 6.0}, with T=4.0 providing the best calibration between soft-label smoothing and class boundary sharpness. Feature weight β was evaluated at {0.1, 0.3, 0.5}, with β=0.3 providing the most stable convergence.

The final configuration is moderately robust to parameter variation. Changing T by ±2.0 affects final accuracy by less than 0.3 percentage points, and changing β from 0.3 to 0.1 reduces accuracy by approximately 0.4 points. The pruning ratios exhibit the strongest sensitivity: reducing κ_1 from 0.85 to 0.75 costs approximately 0.8 points in accuracy while providing only marginal additional FLOP savings, since the first-stage pruning still operates on relatively shallow representations regardless of ratio. The α annealing schedule (0.70→0.50) was adopted from prior distillation literature and was not independently searched.

IV. DISCUSSION AND LIMITATIONS

The central result — a 50%-token student surpassing its dense teacher on CIFAR-100 — challenges the assumption that compression and accuracy trade off monotonically. Three factors contribute: stronger training regularisation through RandAugment, MixUp, and label smoothing, which were absent from teacher training; implicit regularisation from token sparsity; and soft-target supervision providing inter-class similarity structure beyond one-hot labels [6].

Interpretability: Progressive token pruning yields an interpretability benefit at no additional cost. Tokens surviving two rounds of post-stage L_2-norm selection are those the network assigns highest representational magnitude after deep contextual processing. The spatial pattern of retained tokens (Fig. 5) therefore functions as a lightweight forward-pass saliency map, available at inference time without any backward computation — unlike gradient-based methods such as GradCAM. This has practical value in applications such as medical imaging, where explainability is a deployment requirement alongside efficiency.

Theoretical Basis for Implicit Regularisation: The regularisation effect has an information-theoretic interpretation: by constraining predictions to a reduced token set, PTP induces an information bottleneck in which the model must maximise mutual information between the CLS representation and the class label from fewer inputs. This discourages reliance on low-information-density patches such as background regions and uniform textures, and is structurally analogous to stochastic depth and masked autoencoding, both of which are established regularisers. As empirical support, we compute mean prediction entropy $H(p) = -\Sigma\, p_i \log p_i$ on the CIFAR-100 test set. PTP-FAKD produces lower entropy on correctly classified samples (0.33) than the dense teacher (0.41), indicating more confident and concentrated predictions. The naive sparse model shows the highest entropy (0.58), confirming that unstructured single-stage pruning degrades representational quality.

Scalability and Architectural Generalisation: The PTP framework partitions encoder blocks into three equal stages, making it directly applicable to deeper models: ViT-L/16 (24 blocks, stages of 8) and ViT-H/14 (32 blocks, stages of ~11). Since attention cost scales as $O(N^2 \cdot d)$, larger embedding dimensions amplify absolute FLOP savings. For DeiT [7], adaptation is direct given the identical block structure. For Swin, which lacks a global CLS token, average-pooled patch representations would replace CLS-based scoring and stage boundaries would align with Swin's hierarchical stages. Empirical validation on these architectures and at ImageNet scale is identified as the primary future work direction.

Deployment Implications: The 73.5% reduction in attention FLOPs translates directly to reduced key-value cache memory and lower inference latency in edge deployment scenarios — mobile devices, embedded systems, and real-time video pipelines — where quadratic attention is the primary throughput bottleneck. Realising wall-clock improvements in practice requires inference frameworks with variable-length attention kernel support, such as FlashAttention-2.

Limitations: All primary experiments use CIFAR-100. Generalisation to ImageNet-scale training remains unvalidated — the accuracy improvement over the dense teacher may be smaller on larger datasets where overfitting pressure is reduced. Only ViT-B/16 is evaluated, and wall-clock speedup is not benchmarked directly. The naive + feature KD ablation ran for 15 epochs rather than 25, potentially underestimating its best-case performance. The L_2-norm scoring criterion is a heuristic; attention-weight-based scoring may perform better at aggressive token budgets (≤40%). Formal robustness evaluation on corrupted benchmarks and downstream task evaluation on object detection are identified as concrete future work directions.

V. CONCLUSION

We presented PTP-FAKD, a training pipeline combining Progressive Token Pruning with Feature-Aware Knowledge Distillation for efficient Vision Transformers. By deferring and distributing token removal across three encoder stages, PTP ensures that L_2-norm-based pruning decisions are made on contextually enriched hidden representations rather than uninformative embedding-level statistics. FAKD supplements training with CLS token alignment at pruning boundaries, where student-teacher divergence is largest. On CIFAR-100, PTP-FAKD achieves 89.60% top-1 accuracy at 50% token density and 0.265× relative attention FLOPs, surpassing the 87.03% dense teacher by 2.57 percentage points. Controlled ablation establishes that progressive pruning contributes +10.69% over the naive sparse baseline, with FAKD adding a further +0.62%.

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